

AI Readiness Hackathon – KSA Final Submission

Team name: ALARAD

Members name:

- Rawan Hilal Alosaimi — Rahaf Hassan Alkhathlan — Manar Fahad Almarghalani

Solution name: ALARAD — AI-powered predictive radiation safety ecosystem

Contact details: rawanalosaimi1234@gmail.com — rahafalkhathlan@outlook.com — manaralmarghalani@gmail.com

Princess Nourah bint Abdulrahman University – Kingdom of Saudi Arabia

1. ALARAD Overview

1.1 What is the Problem?

Radiology staff are frequently exposed to ionizing radiation. While traditional dosimeters are used, they typically record the dose only after exposure has occurred and do not help prevent it beforehand. Dose data is often fragmented, making it difficult to use for predicting risks or optimizing task distribution among staff.

Millions of healthcare professionals worldwide work in radiation environments every day. Despite wearing dosimeters, they are often alerted only after exposure has already occurred — current radiation monitoring is reactive, not predictive.

1.2 Who are the Users?

Main users: Radiologic Technologists, Medical Physicists, Interventional Radiologists, Nuclear Medicine Staff, Radiation Therapists, and Cath Lab Staff.

Secondary users: Radiation Safety Officers (RSO), Hospital Administrators, and Quality & Safety Departments. Patients indirectly benefit from reduced occupational exposure and improved workflow.

1.3 How Does ALARAD Work?

- Data collection: the smartwatch, connected to an electronic dosimeter, continuously collects real-time exposure data plus contextual information (procedure, duration, location, activity).
- Secure transmission: data is sent securely to the ALARAD mobile application and cloud platform.
- AI analysis: the AI engine analyses real-time and historical data to identify exposure patterns, predict future dose, assess risk levels, and detect abnormal situations before they become hazardous.
- Intelligent recommendations: rotating staff assignments, limiting exposure time, increasing distance, or additional shielding.
- Real-time alerts: if the AI predicts an approach to unsafe exposure limits, the smartwatch sends vibration and visual alerts and notifies the hospital dashboard.
- Hospital dashboard: Radiation Safety Officers and administrators monitor exposure hospital-wide for workforce planning, compliance monitoring, and safety management.
- Continuous learning: as more data is collected, the AI's predictions and recommendations improve over time.

1.4 Where Does AI Come In?

AI does not measure radiation; it analyses data and makes predictions — predicting expected dose if a task continues, identifying the highest-risk individuals, suggesting staff rotation, detecting abnormal patterns, and issuing alerts before regulatory limits are exceeded. AI models used: time-series forecasting, risk prediction, and anomaly detection.

1.5 What Data Does the AI Need?

Exposure data: current dose, cumulative dose, dose rate.

Working data: procedure type, duration, equipment used, department.

User data: job role, work schedule, previous exposure history.

Environmental data: distance from radiation source, shielding availability, room location, scatter radiation level.

1.6 What Makes ALARAD Different?

Traditional Dosimeter	ALARAD
Measures dose	Predicts future exposure
Passive	AI-powered
Alerts after exposure	Warns before dangerous exposure
Individual records	Hospital-wide analytics
No recommendations, no prediction	Provides safety recommendations, predictive risk assessment
Manual reporting, no workflow optimization	Automated reporting, AI-assisted workflow optimization

Traditional dosimeters tell you what happened. ALARAD helps prevent what could happen.

1.7 Expected Impact

- Reduce unnecessary occupational exposure
- Improve compliance with radiation safety standards
- Support hospital decision-making
- Enhance workforce safety

2. Gaps in Existing Solutions

Existing solutions rely on traditional dosimeters, which have the following gaps:

- Passive, not AI-powered — they measure dose rather than predicting future exposure
- Reactive — they alert staff after exposure rather than warning before dangerous exposure occurs
- Fragmented — individual records only, with no hospital-wide analytics
- No decision support — no recommendations, no prediction, manual reporting, and no workflow optimization

ALARAD addresses these gaps directly: AI-driven prediction (time-series forecasting, risk prediction, anomaly detection) closes the passive/reactive gap by detecting abnormal exposure before it becomes hazardous; personalized recommendations close the decision-support gap; and the hospital-wide dashboard closes the fragmentation gap, enabling hospital-wide compliance monitoring and workforce planning.

3. AI Readiness & Mapped Documents

ALARAD mapped onto the seven nodes of the ITU-T Y.3172 AI/ML framework, with the use case and the related policy/regulatory documents for each node.

Node	Use case (ALARAD – radiation safety domain)	Documents (policies, strategies, regulations)
SRC	ALARAD smartwatch connected to an electronic dosimeter worn by radiology staff; collects real-time dose, dose rate, and contextual data (procedure, location, activity).	IAEA GSR Part 3 – Radiation Protection and Safety of Radiation Sources NRRC-R-01-SR03 – QC Program for Medical Radiological Equipment
C	ALARAD mobile application and cloud platform receiving data securely transmitted from the smartwatch.	NCA Operational Technology Cybersecurity Controls Methodology and Mapping Annex (OTCC-1:2022)
PP	Pre-processing on the mobile app / cloud platform: cleaning and structuring exposure, working, user, and environmental data before model input.	SDAIA/NDMO – National Data Management and Personal Data Protection Standards
M	Time-series forecasting (future dose), risk prediction (risk level), anomaly detection (abnormal exposure patterns).	ICRP Publication 103 – The 2007 Recommendations of the ICRP JKSUS – Occupational radiation exposure among diagnostic radiology workers in Saudi MOH hospitals (5-year study)
P — Human-in-the-loop	Model outputs (predicted dose, risk level, anomaly flags) are checked against dose limits; high-risk alerts are reviewed by a Radiation Safety Officer before escalation.	NRRC-R-01-SR02 – Compliance with Dose Limits NRRC – Nuclear and Radiological Regulatory Legal Framework IAEA SSG-46 – Radiation Protection and Safety in Medical Uses of Ionizing Radiation SDAIA/NDMO Personal Data Protection Standards (health data handling, purpose limitation)
D	Recommendations and alerts routed to the smartwatch (individual) and the hospital dashboard (department-wide).	NCA OTCC-1:2022 (secure distribution across hospital OT/IT systems)
SINK	The smartwatch (end-user alert) and the hospital dashboard (Radiation Safety Officer / hospital administration).	ITU – AI Ready: Analysis Towards a Standardized Readiness Framework, Report 2.0 (Jan 2026)

Knowledge Base – Reference Links

1. SDAIA/NDMO – National Data Management and Personal Data Protection Standards: https://sdaia.gov.sa/ndmo/Files/PoliciesEn001.pdf?utm_source=chatgpt.com
2. Occupational radiation exposure among diagnostic radiology workers in Saudi MOH hospitals (JKSUS, 5-year study): <https://jksus.org/occupational-radiation-exposure-among-diagnostic-radiology-workers-in-the-saudi-ministry-of-health-hospitals-and-medical-centers-a-five-year-national-retrospective-study/>
3. NCA – Operational Technology Cybersecurity Controls Methodology and Mapping Annex (OTCC-1:2022): https://cdn.nca.gov.sa/api/files/public/upload/071d52fc-014b-4f15-84ce-1289f3f5c3a9_Operational-Technology-Cybersecurity-Controls-Methodology-and-Mapping-Annex.pdf
4. NRRC – Nuclear Regulations and Executive Regulations (Legal Framework): <https://nrcc.gov.sa/en/legal-framework/nuclear-regulations-and-executive-regulations/>
5. IAEA GSR Part 3 – Radiation Protection and Safety of Radiation Sources: https://www-pub.iaea.org/MTCD/Publications/PDF/Pub1578_web-57265295.pdf
6. IAEA SSG-46 – Radiation Protection and Safety in Medical Uses of Ionizing Radiation: <https://www.iaea.org/publications/11102/radiation-protection-and-safety-in-medical-uses-of-ionizing-radiation>
7. AI Readiness Hackathon – Kingdom of Saudi Arabia (ITU AI for Good, official page): <https://aiforgood.itu.int/event/ai-readiness-hackathon-kingdom-of-saudi-arabia/>
8. ICRP Publication 103 – Free Extract: [https://www.icrp.org/docs/icrp_publication_103-annals_of_the_icrp_37\(2-4\)-free_extract.pdf](https://www.icrp.org/docs/icrp_publication_103-annals_of_the_icrp_37(2-4)-free_extract.pdf)
9. ITU – AI Ready: Analysis Towards a Standardized Readiness Framework, Report 2.0 (Jan 2026): https://www.itu.int/dms_pub/itu-t/opb/ai4g/T-AI4G-AI4GOOD-2025-6-PDF-E.pdf
10. NRRC-R-01-SR02 – Compliance with Dose Limits: <https://istitlaa.ncc.gov.sa/ar/energy/nrcc/compliancewithdoeslimits/Documents/Compliance%20with%20Dose%20Limits-NRRC-R-01-SR02.pdf>
11. NRRC-R-01-SR03 – Establishment and Implementation of QC Program for Medical Radiological Equipment: [https://istitlaa.ncc.gov.sa/en/energy/nrcc/establishmentandimplementationofqcprogram/Documents/Establishment%20and%20Implementation%20of%20Quality%20Control%20\(QC\)%20Program%20for%20Medical%20Radiological%20Equipment-2025_Istitlaa.pdf](https://istitlaa.ncc.gov.sa/en/energy/nrcc/establishmentandimplementationofqcprogram/Documents/Establishment%20and%20Implementation%20of%20Quality%20Control%20(QC)%20Program%20for%20Medical%20Radiological%20Equipment-2025_Istitlaa.pdf)

4. Evaluation Scenarios

Scenario A — Intended operation

Step 1: ALARAD is deployed across multiple radiology departments in a Saudi hospital and becomes widely used by interventional radiologists and cath lab staff during daily procedures.

Step 2: During a routine procedure, a radiation leak occurs in a treatment room (e.g. a shielding failure or equipment malfunction), producing an abnormal, unplanned radiation source. ALARAD's anomaly detection model identifies the abnormal exposure pattern in real time, distinguishing it from normal fluctuation.

Step 3: The system triggers a real-time alert to the affected staff member's smartwatch and escalates the event to the hospital dashboard, notifying the Radiation Safety Officer. The event is assessed against the applicable dose limits and radiation-safety procedures, including NRRC-R-01-SR02 where applicable. The hospital radiation-safety department documents and investigates the incident, and any required regulatory notification is handled according to the applicable NRRC reporting requirements and relevant medical-use radiation-safety guidance.

Scenario B — Controversy: misuse of exposure data

Step 1: ALARAD is deployed hospital-wide and staff come to trust its alerts and recommendations as part of daily practice.

Step 2: After several months, it is noticed by the hospital's quality and safety department that individual dose and alert history is being pulled into staff performance reviews — for example, staff who trigger frequent “elevated exposure” alerts are being flagged in annual evaluations, even when the elevated readings result from unavoidable procedure types (e.g. high-volume interventional cases) rather than staff negligence.

Step 3: This escalates the case from an operational monitoring issue to a data-governance and fairness concern. The knowledge base should trigger purpose-limitation and data-governance safeguards: exposure and alert data collected for radiation-safety purposes should not automatically be reused for unrelated staff-performance decisions. Any proposed secondary use should be assessed against the applicable personal-data requirements, lawful basis, transparency, access controls, and institutional policies. The Policy (P) node's human-in-the-loop requirement is extended so that uses of ALARAD data outside radiation-safety response are reviewed by the appropriate data-protection and compliance personnel. Where the secondary use is not justified, the hospital should stop or remediate it and notify affected staff as required. This demonstrates that ALARAD's policy mapping governs not only dose thresholds, but also responsible use of the data generated by the system.